The water-path detectives

Teacher guide and worked answers

Explain the root-stem-leaf water pathway and use dye evidence to identify internal transport in a cut shoot.

40 minutes. Use Lesson.html for interactive teaching or Lesson.pptx for editable slides. Slides.pdf is the static alternative. Select one pupil response route and print the shared evidence it requires. Detailed answers below are for staff.

Scheme of work

_passsb/inputs/Build SOW 2026-2027.xlsx | BUILD Weekly - Summer | C35

Explain how water travels through a plant.

Chronological continuation of the supplied SOW. This lesson develops or reviews the named outcome; it does not claim an accreditation award.

Prior learning

Roots absorb water and minerals.

Stems contain transport pathways.

Success evidence

Trace soil water through roots and stem to leaves.

Explain what coloured internal pathways support.

State a limit of a rootless demonstration.

Equipment

Printed shared evidence and one chosen pupil route; pencil; optional ruler; projector or offline HTML.

Optional adult-prepared cut white-flower shoots, stable plastic cups, dilute food colouring, plain-water comparison and tray; prepared stem sections.

Preparation

Print the shared evidence, one response route and exit page. Routes are alternatives, not a workload ladder.

Read the worked answers. Prepare an oral, pointing or adult-scribed route where useful.

Prepare any coloured-water shoots and plain-water comparison well before the lesson; real staining may take hours or longer and can vary. The supplied constructed observations allow full teaching without waiting.

Only an adult cuts stems or sections using the school’s usual safe method, outside pupil handling. Keep a comparison in plain water and prepare labelled observation cards. A cut celery stalk is a leaf stalk, not a complete stem; this pack uses a cut flower stem.

Practical care

Use adult-selected non-toxic plant material and small amounts of dilute food colouring; nothing is eaten or drunk.

Adults prepare all cuts; pupils do not use blades. Cover surfaces, use stable cups in trays, wipe spills and wash hands.

Keep liquids away from electrical devices. No pressurised or heated apparatus is needed.

Ready to teach alternative

All core reasoning uses the supplied evidence and can be completed in 40 minutes without a live practical.

Teaching sequence

Slide	Stage	Minutes	Focus
1	Opening	0-2	How can water reach a leaf above the soil?
2	Retrieval	2-5	Retrieve the connected parts
3	I do	5-12	I trace the rooted-plant route
4	I do	Within stage	I mark water in a cut shoot
5	I do	Within stage	I inspect the interior evidence
6	We do	12-16	Sequence the evidence before explaining
7	We do	16-20	Which claim reaches too far?
8	Hinge	20-23	What does an internal red path support?
9	You do	23-25	Choose your science route
10	You do	25-35	Make your own evidence record
11	You do	Within stage	Check what the evidence can show
12	Review	35-38	Lundy loop: improve the shared account
13	Review	Within stage	Keep the useful change
14	Exit	38-40	Three final checks

Times are a suggested 40-minute route. Slides marked Within stage share the preceding stage allocation. The optional HTML timer starts only when selected.

1 Opening How can water reach a leaf above the soil?

Distinguish the rooted-plant explanation from the rootless demonstration. Neither pupils nor the lesson need to wait for colour to develop.

2 Retrieval Retrieve the connected parts

1 roots. 2 stem. 3 no; roots absorb water and dissolved mineral nutrients, not chunks of soil.

3 I do I trace the rooted-plant route

Think aloud: “I begin with water in soil, enter a root and follow xylem through the stem to a leaf. Some water evaporates and leaves as vapour.” Evaporation from leaves helps pull water upward; a straw-sucking mouth is not the mechanism.

4 I do I mark water in a cut shoot

Shares I do time. Think aloud: “I can investigate movement in this shoot, but I cannot claim to have shown root uptake because there are no roots.” Food colouring marks some pathways; it is not normal plant food.

5 I do I inspect the interior evidence

Shares I do time. Think aloud: “Inside colour supports transport through internal pathways. A coloured outer surface alone would be weaker evidence.” Xylem name is helpful but not required for supported core explanation.

6 We do Sequence the evidence before explaining

W1: B then C then D. Use evidence reader, sequencer and sceptic roles. Ask what the plain-water comparison adds: colour comes from the dye condition. Records are constructed; no universal 2-hour or 24-hour rate is implied.

7 We do Which claim reaches too far?

W2: first supported; second unsupported because no roots. F also blocks “every missing drop entered the shoot”, because evaporation can reduce water level.

Reveal: Explain the demonstrated part of the route and state what was not tested.

8 Hinge What does an internal red path support?

Correct first option. Revisit the section diagram if pupils treat the entire stem as one hollow pipe.

A. Coloured water moved through internal pathways. - The tracer appears within the cut section.

B. The demonstration directly observed root absorption. - There are no roots in this setup.

C. The dye is the plant’s natural food. - It is a tracer; green plants make their own sugars.

9 You do Choose your science route

Set ONE route. Require a connected route and a claim limited to the evidence. A diagram with correctly directed arrows can replace prose. Do not require detailed transpiration physics or a perfect anatomical section.

10 You do Make your own evidence record

Require a connected route and a claim limited to the evidence. A diagram with correctly directed arrows can replace prose. Do not require detailed transpiration physics or a perfect anatomical section.

11 You do Check what the evidence can show

Shares independent time. Ask for a reason, not just a correct word. Require a connected route and a claim limited to the evidence. A diagram with correctly directed arrows can replace prose. Do not require detailed transpiration physics or a perfect anatomical section.

12 Review Lundy loop: improve the shared account

Listen to a selected pupil revision and visibly adopt a scientifically accurate contribution. Offer private, written or partner feedback.

Reveal: The rootless shoot supports internal transport, while the full root pathway is a separate scientific model.

13 Review Keep the useful change

Shares review time. Name how a pupil contribution changed a label, method or conclusion. Do not rank pupils.

Reveal: The rootless shoot supports internal transport, while the full root pathway is a separate scientific model.

14 Exit Three final checks

Collect brief individual evidence. Use the answer key and re-teach the specified misconception if needed.

Answers and assessment

Retrieval 1-3

1 roots. 2 stem. 3 no; water with dissolved minerals is absorbed.

We do W1-W2

W1 B setup, C internal stem colour, D later petal colour. E provides a plain-water comparison. W2 root uptake was not directly tested; no roots present. F shows evaporation as another cause of lost water.

Hinge

First option correct. Internal tracer supports internal movement; no roots were present; dye is not natural plant food.

S1-S5

S1 roots, stem. S2 inside. S3 plain water without red dye. S4 no. S5 coloured water moved through internal pathways towards upper parts.

M1-M4

M1 soil → roots → stem/xylem → leaves, with some water exiting as vapour if added. M2 tracer inside stem and later petals supports movement within the shoot to upper parts. M3 no red dye appears without the dye condition, helping distinguish tracer staining from a pre-existing colour. M4 roots were absent, so direct root uptake was not observed; F shows open water can evaporate without a plant.

T1-T4

T1 inside staining contradicts outside-only movement. T2 the shoots were cut and rootless. T3 evaporation from the exposed water surface. T4 e.g. the coloured water produced internal stained paths and later petal colour, supporting movement through this cut shoot. It did not directly test root uptake or show that every drop lost from the cup entered the plant.

Exit E1-E3

E1 roots. E2 yes, through xylem pathways. E3 no.

R1-R2

Responses vary. Credit a scientific revision, evidence-based defence or relevant next question. A private or oral response has equal value.

Teaching and assessment notes

40 minutes: opening 2, retrieval 3, I do 7, We do 8, hinge 3, independent 12, review 3, exit 2. Zero-minute slides share the named stage time.

BUILD uses the supplied Year 3 science anchor with age-respectful contexts for older pupils. Supported, standard and stretch are selectable routes within the pathway.

Interactive We do: predict first, test against the controlled SVG model or supplied evidence, then explain or revise. Shared W tasks offer the equivalent paper discussion. Animation is a teaching representation, not filmed experimental evidence.

Keep constructed evidence separate from actual class observations. Do not invent measurements or require internet research.

Secure core: roots → stem → leaves with internal transport, plus recognition that the demonstration lacks roots. Re-teach the limit by physically pointing to what is and is not in the apparatus.

Access and responsive teaching

Supported

Short choices, word bank and pointing or adult-scribed reasons; same core scientific goal.

Standard

Make a short evidence-based explanation or record using the supplied information.

Stretch

Apply the core idea to a changed case, evaluate evidence or identify a limit.

Regulation

Preview the sequence. Offer seated observation, quiet paired rehearsal, private feedback and a pass from public sharing. Routes respond to current access needs, not a diagnosis.

Misconceptions to check

Water mainly travels up the outside of the stem.

The dye comparison can reveal internal water-carrying pathways.

Check: Ask where the adult-cut section shows colour.

A cut shoot proves how roots absorb water.

The cut shoot has no roots; it gives evidence about transport above the cut.

Check: Ask whether roots are present in the setup.

Any falling water level proves every lost drop entered the plant.

Water can also evaporate from a container.

Check: Compare a cup without a plant.

Word	Meaning
transport	Movement of a substance from one place to another.
xylem	Plant pathways that carry water and dissolved minerals.
tracer	A marker used to help follow movement.
water vapour	Water in its gas state.

Scientific references

Department for Education: science programmes of study

Checked 8 September 2026. Year 3 plants, rocks, light and forces underpin the SOW. Accessible teaching models are adapted to the named lesson objective.

Science and Plants for Schools: water transport

Checked 8 September 2026. Xylem transports water and dissolved minerals; evaporation from leaves contributes to pulling water upwards. Phloem sugar transport is distinct.

Science and Plants for Schools: how does water travel through a plant?

Checked 8 September 2026. Dye evidence can reveal internal water pathways. Our cut-shoot demonstration is explicitly not a complete rooted plant.

Science and Plants for Schools: plant biology animations

Checked 8 September 2026. Scientific background for our original simplified diagrams: photosynthesis, growth and transport. No external animation is required or embedded.